

University Of Al Maarif

College of Engineering

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Database Design & Implementation for Football Management System

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First stage | Database System Course | Academic Year: 2025-2026

1. Objective:

- The project is design an extensive relational database that are models the structure for a football management system
- Represent all keys entity and their relationships by using entity relationship diagram (ERD)
- Convert the ERD into functional relational **schema** with primary and foreign keys (that put in the end of it)
- Implement the schema by using SQL queries by using MySQL workbench
- Develop a set of SQL queries to demonstrate the system ability to answer the user questions
- Check the system supports operations like tracing the match results and record training sessions and so on at all

2. System Description

The Football management System is designed to manage semi complete lifecycle of data that covers the following keys domain:

2.1 Person

By using Generalization (circle junction) and Specialization (triangular junction) patterns that are coming from **person** entity gives a multiple attributes (Name as composite attribute. Birth date, Nationality, Email as simple attribute. Phone numbers as multivalued attribute) shared by player and coach and referee.

- ✓ Player : includes player name, market value, position, shirt number, and total goals (driven)
- ✓ Coach : includes coach name, license level, tactical style, experience, and languages (multivalued)
- ✓ Referee : includes referee name, badge type, international status, and experience.

2.2 Club

The club is given with its official name and founded year with a composite attribute of address (country, city, street). A club can own multiple players and employ multiple coaches sync at the same time

2.3 Match

Any match will recorded the details like the match statistics and participating home / away clubs, match date and time, weather condition, round, season and so on

2.4 Training Sessions

The coach will organize training sessions for players by a session type, date, and duration. Players participate in these sessions, establishing a many-to-many relationship between players and training sessions

3. Methodology

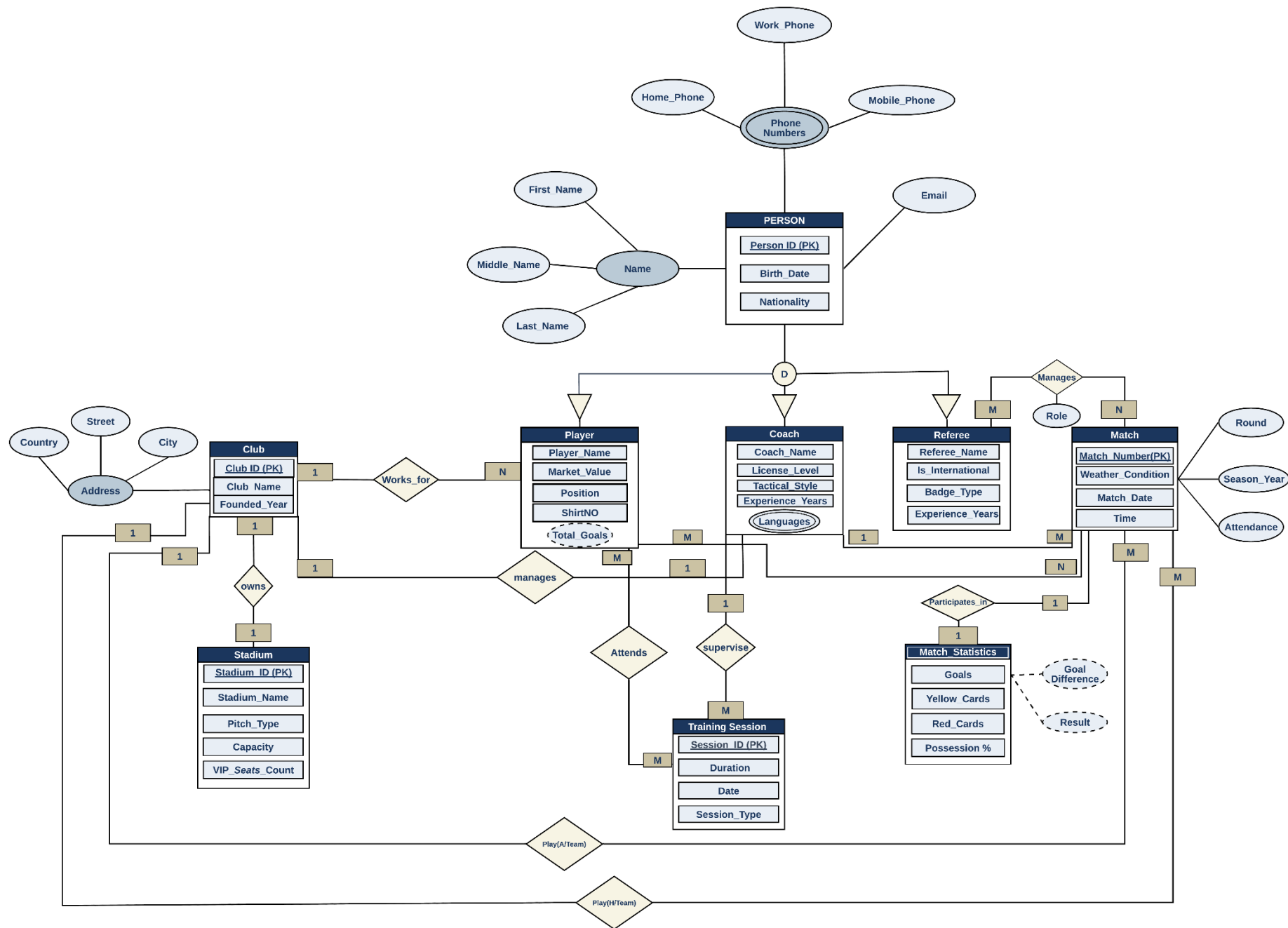
3.1 Concept of EDR

- Entities: Person, Player, Coach, Referee, Club, Match, Training Session, Stadium
- Relationships & cardinalities: (1:1, 1:M, M:N)
- Attributes: including composite (Name) and multivalued (Phone) and driven attributes

Specialization: Person is the generalized entity for Coach and Player and Referee

3.2 Schema

- Strong entity became table with a PK firstly and FK at the end if needed
- Simple attribute will be inserted into their related entities
- Multivalued attribute (phone numbers) inserted into new separated table (phone)
- Composite attribute we will take their simple attribute
- Driven attribute deal as a simple attribute
- Entity relation will be inserted into new separated table (role)
- M:N relationship were resolved using junction tables

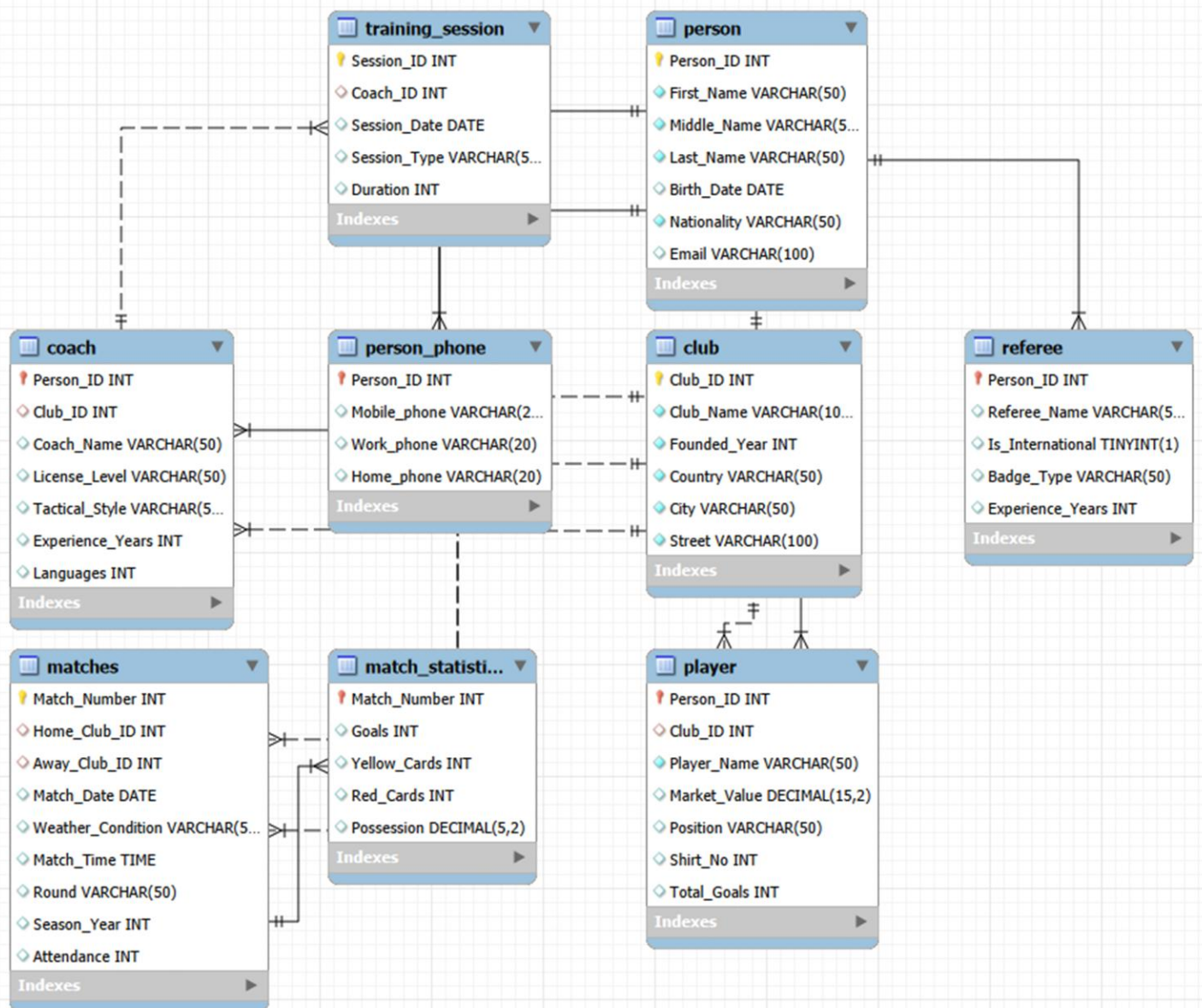


4. SQL Implementation

Schema was implemented in MySQL workbench using DDL (CREATE TABLE) statements. Constraints including (PK, FK, NOT NULL) and CHECK if it is applied or not to enforce data integrity. Our sample data was inserted by using DML (INSERT INTO) statements

5. Query Development & Testing

A set of SQL SELECT * FROM queries was developed to see the system. Queries include JOINS and other functions across multiple tables of our system that we need to add and conditional filtering to answer the user realistic football management questions



Person_ID	First_Name	Middle_Name	Last_Name	Birth_Date	Nationality	Email
1	Carlo	Michelangelo	Ancelotti	1959-06-10	Italian	carlo.ancelotti@football.com
2	Xavi	Creus	Hernandez	1980-01-25	Spanish	xavi.hernandez@football.com
3	Diego	Pablo	Simeone	1970-04-28	Argentine	diego.simeone@football.com
4	Quique	Sanchez	Flores	1965-02-05	Spanish	quique.flores@football.com
5	Ruben	Vegas	Baraja	1975-07-11	Spanish	ruben.baraja@football.com
6	Marcelino	Garcia	Torral	1965-08-14	Spanish	marcelino.torral@football.com
7	Imanol	Barrenetxea	Alguacil	1971-07-04	Spanish	imanol.alguacil@football.com
8	Manuel	Luis	Pellegrini	1953-09-16	Chilean	manuel.pellegrini@football.com
9	Ernesto	Tejedor	Valverde	1964-02-09	Spanish	ernesto.valverde@football.com
10	Jose	Jimenez	Bordalas	1964-03-05	Spanish	jose.bordalas@football.com
11	Rafael	Maudes	Benitez	1960-04-16	Spanish	rafael.benitez@football.com
12	Jagoba	Elustondo	Arrasate	1978-04-22	Spanish	jagoba.arrasate@football.com
13	Luis	Plaza	Garcia	1980-12-01	Spanish	luis.garcia@football.com
14	Javier	Onaindia	Aguirre	1958-12-01	Mexican	javier.aguirre@football.com
15	Garcia	Javier	Pimienta	1974-08-03	Spanish	garcia.pimienta@football.com
16	Francisco	Javier	Rodriguez	1978-06-17	Spanish	francisco.rodriguez@football.com
17	Michel	Angel	Sanchez	1975-10-30	Spanish	michel.sanchez@football.com
18	Mauricio	Andres	Pellegrino	1971-10-05	Argentine	mauricio.pellegrino@football.com
19	Alexander	Jesus	Medina	1978-08-08	Uruguayan	alexander.medina@football.com
20	Gaizka	Aguirre	Garitano	1975-07-09	Spanish	gaizka.garitano@football.com
21	Vinicius	Jose	Junior	2000-07-12	Brazil	vini.jr@football.com
22	Jude	Victor	Bellingham	2003-06-29	England	jude.b@football.com
23	Robert	Guy	Lewando...	1988-08-21	Poland	robert.l@football.com
24	Pedri	Gonzalez	Lopez	2002-11-25	Spain	pedri.g@football.com
25	Antoine	Jean	Griezmann	1991-03-21	France	antoine.g@football.com
26	Jan	Nic	Oblak	1993-01-07	Slovenia	jan.o@football.com
27	Youssef	Ali	En-Nesyri	1997-06-01	Morocco	youssef.e@football.com
28	Jose	Luis	Gaya	1995-05-25	Spain	jose.g@football.com
29	Gerard	Balaguero	Moreno	1992-04-07	Spain	gerard.m@football.com
30	Takefusa	Ken	Kubo	2001-06-04	Japan	take.kubo@football.com

Person_ID	Mobile_phone	Work_phone	Home_phone
1	+34-611-111111	+34-911-111111	+34-910-222222
2	+34-622-222222	+34-922-222222	+34-920-333333
3	+34-633-333333	+34-933-333333	+34-930-444444
4	+34-644-444444	+34-944-444444	+34-940-555555
5	+34-655-555555	+34-955-555555	+34-950-666666
6	+34-666-666666	+34-966-666666	+34-960-777777
7	+34-677-777777	+34-977-777777	+34-970-888888
8	+34-688-888888	+34-988-888888	+34-980-999999

Person_ID	Referee_Name	Is_International	Badge_Type	Experience_Years
9	Ernesto Valverde	1	FIFA	15
10	Jose Bordalas	1	FIFA	12
11	Rafael Benitez	1	FIFA	10
12	Jagoba Arrasate	1	FIFA	8
13	Luis Garcia	0	National	5

Person_ID	Club_ID	Coach_Name	License_Level	Tactical_Style	Experience_Years	Languages
1	1	Carlo Ancelotti	Pro	Balanced	30	0
2	2	Xavi Hernandez	Pro	Possession	5	0
3	3	Diego Simeone	Pro	Defensive	15	0
4	4	Quique Flores	Pro	Counter Attack	20	0
5	5	Ruben Baraja	A	Balanced	10	0
6	6	Marcelino Torral	Pro	Attacking	22	0
7	7	Imanol Alguacil	Pro	Possession	12	0
8	8	Manuel Pellegrini	Pro	Attacking	35	0

Club_ID	Club_Name	Founded_Year	Country	City	Street
1	Barcelona	1899	Spain	Barcelona	Aristides Maillol
2	Atletico Madrid	1903	Spain	Madrid	Av. Luis Aragones
3	Sevilla	1890	Spain	Seville	Calle Sevilla
4	Valencia	1919	Spain	Valencia	Av. Suecia
5	Villarreal	1923	Spain	Villarreal	Calle Blasco
6	Real Sociedad	1909	Spain	San Sebastian	Anoeta Pasealekua
7	Real Betis	1907	Spain	Seville	Av. Heliopolis
8	Athletic Bilbao	1898	Spain	Bilbao	Rafael Moreno
9	Getafe	1983	Spain	Getafe	Teresa de Calcuta
10	Celta Vigo	1923	Spain	Vigo	Balaidos
11	Osasuna	1920	Spain	Pamplona	Sadar
12	Alaves	1921	Spain	Vitoria	Paseo Cervantes
13	Mallorca	1916	Spain	Palma	Cami dels Reis
14	Las Palmas	1949	Spain	Las Palmas	Fondos de Segura
15	Rayo Vallecano	1924	Spain	Madrid	Payaso Fofo
16	Girona	1930	Spain	Girona	Montilivi
17	Cadiz	1910	Spain	Cadiz	Plaza Madrid
18	Granada	1931	Spain	Granada	Pintor Manuel
19	Almeria	1989	Spain	Almeria	Calle Belladona
20	Real Madrid	1902	Spain	Madrid	Padre Damian

Session_ID	Coach_ID	Session_Date	Session_Type	Duration
1	1	2023-10-25	Tactical	90
2	1	2023-10-26	Physical	120
3	1	2023-10-27	Recovery	60
4	1	2023-11-01	Tactical	90
5	1	2023-11-02	Shooting	75
6	1	2023-11-03	Physical	110
7	1	2023-11-08	Tactical	90
8	1	2023-11-09	Set Pieces	60
9	1	2023-11-10	Recovery	45
10	1	2023-11-15	Tactical	90
11	1	2023-11-16	Physical	100
12	1	2023-11-17	Shooting	80
13	1	2023-11-22	Tactical	90
14	1	2023-11-23	Set Pieces	60
15	1	2023-11-24	Recovery	50
16	1	2023-11-29	Tactical	90
17	1	2023-11-30	Physical	120
18	1	2023-12-01	Shooting	70
19	1	2023-12-06	Tactical	90
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Match_Number	Home_Club_ID	Away_Club_ID	Match_Date	Weather_Condition	Match_Time	Round	Season_Year	Attendance
1	1	2	2023-10-28	Clear	16:15:00	Round 11	2023	80000
2	3	1	2023-09-24	Rainy	21:00:00	Round 6	2023	65000
3	1	4	2024-02-25	Cloudy	21:00:00	Round 26	2023	75000
4	5	2	2023-12-16	Sunny	21:00:00	Round 17	2023	45000
5	6	3	2024-04-01	Rainy	21:00:00	Round 30	2023	20000
6	7	9	2023-09-30	Clear	21:00:00	Round 8	2023	38000
7	8	4	2023-11-12	Windy	18:30:00	Round 13	2023	55000
8	2	3	2023-12-03	Sunny	21:00:00	Round 15	2023	90000
9	10	1	2024-02-01	Cloudy	21:00:00	Round 20	2023	15000
10	1	6	2023-12-17	Rainy	21:00:00	Round 17	2023	70000
11	11	1	2023-08-25	Clear	21:30:00	Round 3	2023	25000
12	12	2	2023-09-03	Sunny	21:00:00	Round 4	2023	22000
13	1	13	2024-05-15	Clear	21:00:00	Round 36	2023	72000
14	14	1	2024-04-13	Cloudy	18:30:00	Round 31	2023	21000
15	1	15	2023-09-27	Rainy	19:00:00	Round 7	2023	68000
16	16	1	2024-02-18	Sunny	14:00:00	Round 25	2023	14000
17	17	1	2023-09-30	Clear	18:30:00	Round 8	2023	13000
18	1	18	2024-05-04	Cloudy	16:15:00	Round 34	2023	74000
19	1	19	2023-12-02	Rainy	18:30:00	Round 15	2023	71000
20	20	1	2023-08-19	Sunny	19:30:00	Round 2	2023	14500

Match_Number	Goals	Yellow_Cards	Red_Cards	Possession
1	3	4	0	52.50
2	4	6	1	48.00
3	1	2	0	60.00
4	2	3	0	55.50
5	3	5	0	45.00
6	2	4	0	50.00
7	1	7	1	49.50
8	1	3	0	62.00
9	2	5	0	65.50
10	5	2	0	58.00
11	1	4	0	61.00
12	3	5	1	70.50
13	4	1	0	66.00
14	1	3	0	57.00
15	2	2	0	64.50
16	2	6	0	53.00
17	3	4	1	51.50
18	3	2	0	68.00
19	2	3	0	63.00
20	4	5	0	59.00

6. References

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2. Draw.io (diagrams.net) Used for ERD and schema diagram creation
<https://app.diagrams.net/>
3. Lucidchart Used as assisted for ERD [Lucidchart](https://www.lucidchart.com/pages)
<https://www.lucidchart.com/pages>
4. MySQL workbench 8.0 program